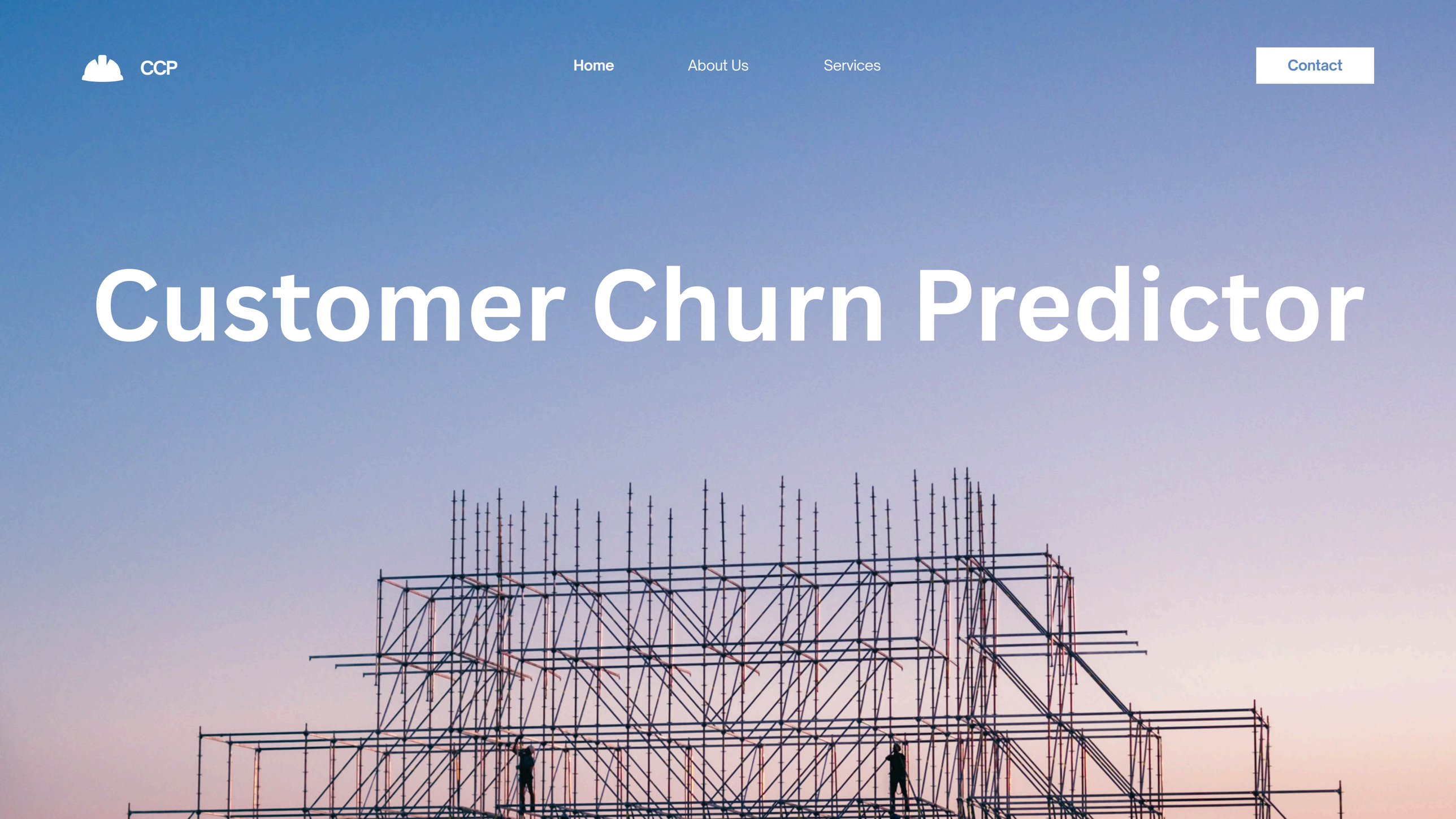
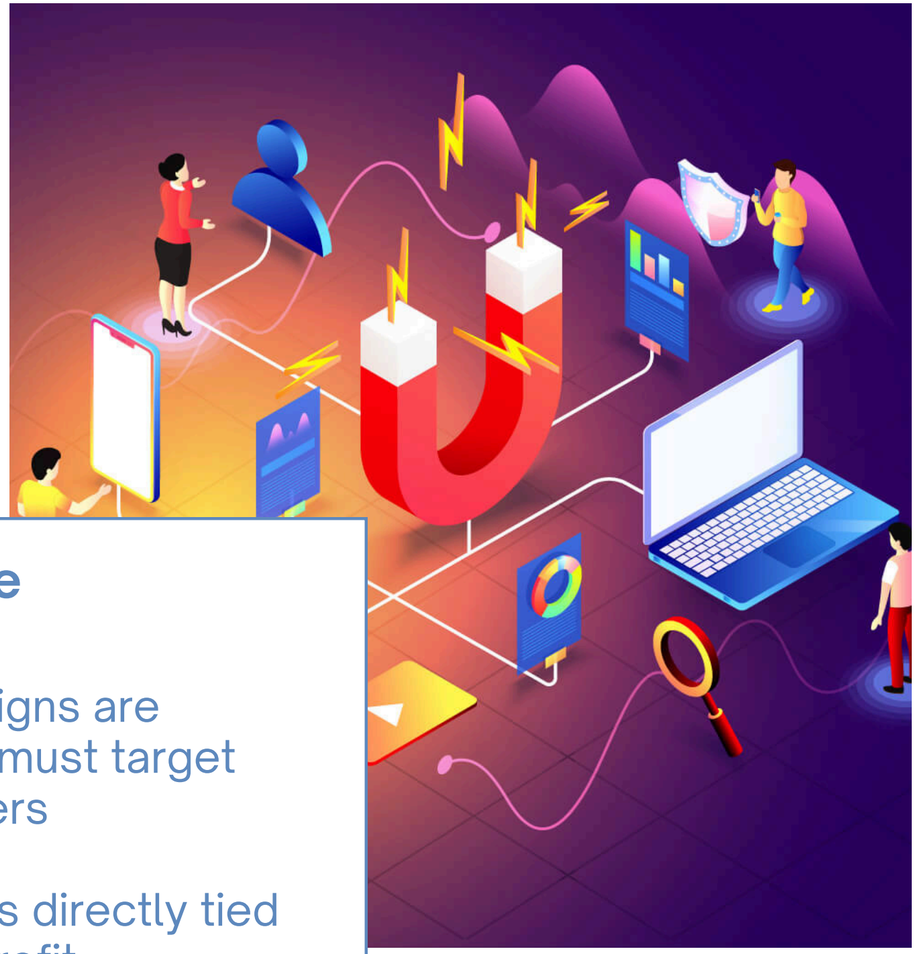
Customer Churn Predictor



Why does Churn Matter?



Business Problem: Customer churn reduces recurring revenue and increases acquisition cost.

Goal: Predict which customers will churn early, so the business can intervene efficiently.

What success looks like

- Rank customers by churn risk (probability)
- Convert risk into a clear action: Intervene vs No action
- Choose interventions based on ROI, not guesswork

Why this is valuable

- Retention campaigns are expensive → we must target the right customers
- Churn reduction is directly tied to revenue and profit

Data + Target Definition

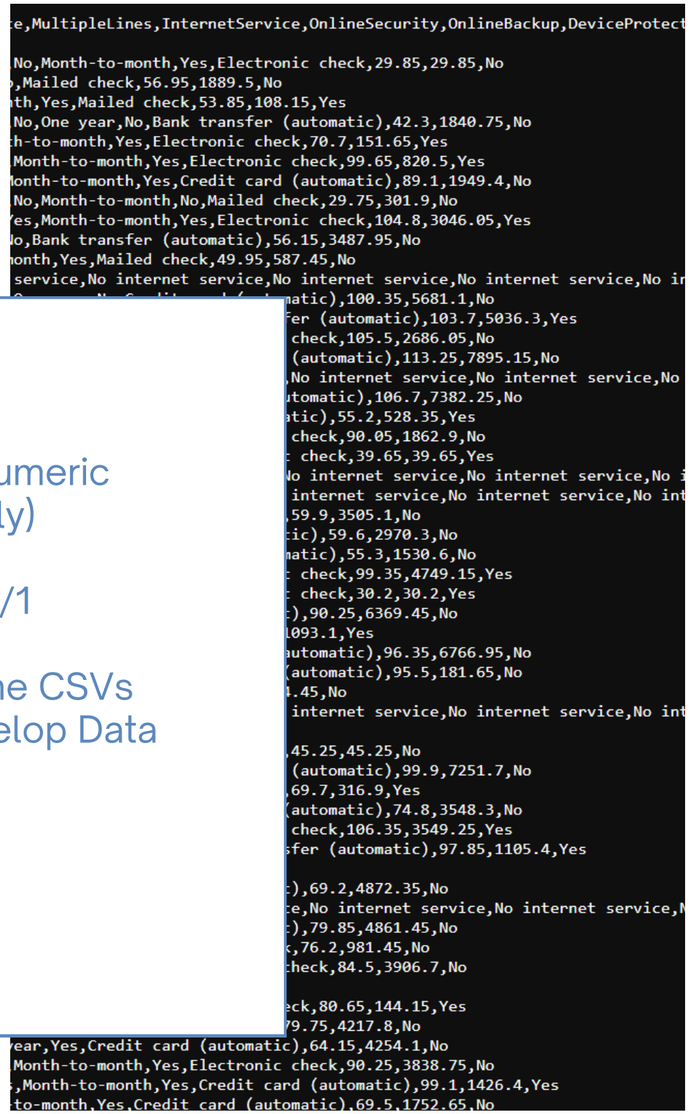
Target (label) Churn: 1 = churned, 0 = stayed

Key feature groups

- Customer: gender, SeniorCitizen, Partner, Dependents
- Services: PhoneService, InternetService, OnlineSecurity, TechSupport, StreamingTV/Movies
- Contract/Billing: Contract, PaperlessBilling, PaymentMethod
- Charges: tenure, MonthlyCharges, TotalCharges

Data cleaning highlights

- Converted TotalCharges → numeric (handled blanks/missing safely)
- Standardized churn label to 0/1
- Once cleaned, we can take the CSVs to import into Tableau to develop Data Visualization Dashboards



Dataset: Telco Customer Churn (customer demographics + services + billing + churn label)

Pipeline + System Architecture (End-to-End)

```
Telco Dataset
    ↓
 Cleaning & Feature Engineering (Pandas)
    ↓
rocessing Pipeline (Impute + Scale + OneHotEn
    ↓
sification Models (LogReg, RF)
    ↓
n Probability Scores (0–1)
    ↓
 Segmentation (Buckets + Deciles)
    ↓
Based Threshold Decisioning
    ↓
API + Streamlit + Monitoring + Tableau Dashbo
```

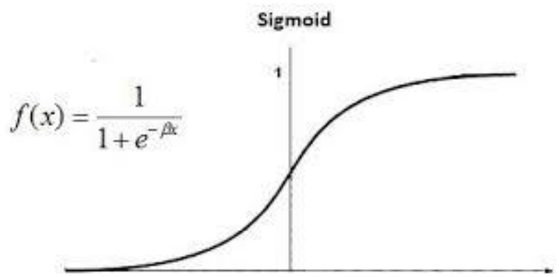
Reproducible ML pipeline (scikit-learn)

- Numeric: median imputation + standard scaling
- Categorical: mode imputation + one-hot encoding
- Implemented via Pipeline + ColumnTransformer (prevents train/inference mismatch)

Architecture

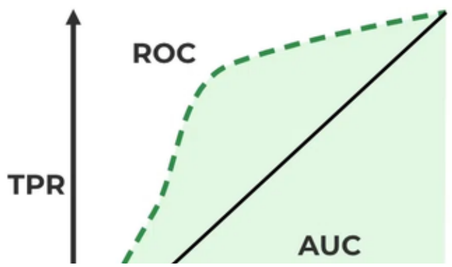
- Training/Evaluation scripts → saves model artifact + metrics
- FastAPI → /predict, /predict_batch, /health + prediction logging
- Streamlit UI → product-style dashboard for users
- Monitoring → Evidently drift report
- Tableau → stakeholder dashboards from exported CSVs

Model Training + Selection



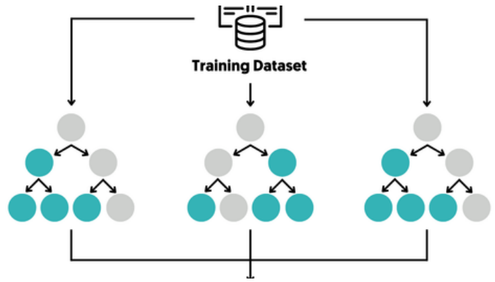
Models compared

- Logistic Regression (interpretable baseline)
- Random Forest (nonlinear candidate)



Selection metric

- Primary: ROC-AUC (ranking quality across thresholds)
- Also tracked: PR-AUC (better signal for imbalanced churn class)



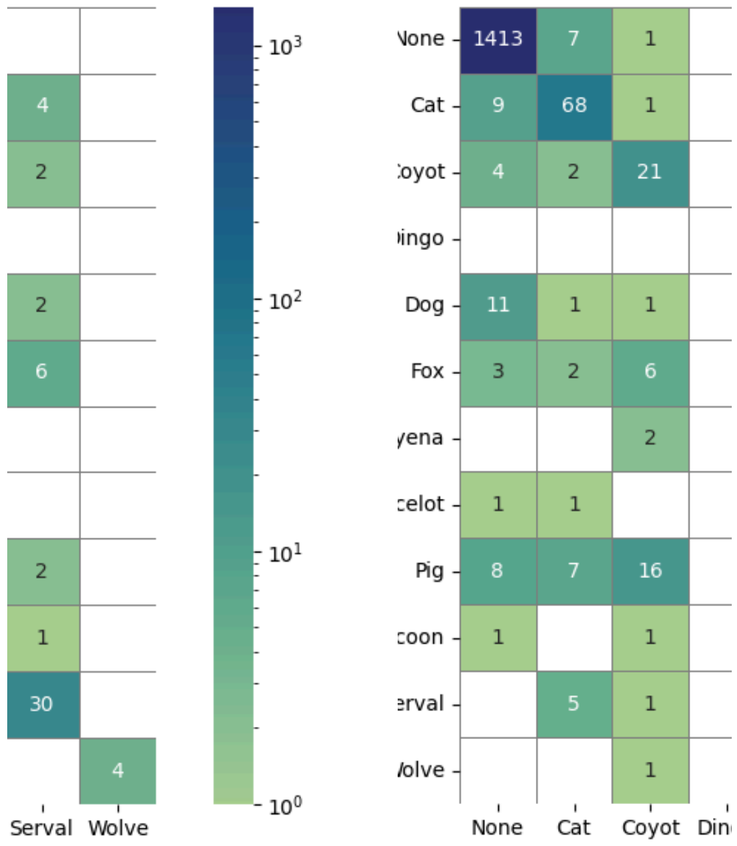
Results (holdout test split)

- Logistic Regression: ROC-AUC 0.8414, PR-AUC 0.6324 ✅ best
- Random Forest: ROC-AUC 0.8224, PR-AUC 0.6135

Final model chosen: Logistic Regression

Reason: best ranking performance + simpler deployment + easier interpretation.

Evaluation: Confusion Matrix + Threshold Tradeoffs



Confusion matrix at threshold = 0.50

- TN=746, FP=289
- FN=81, TP=293

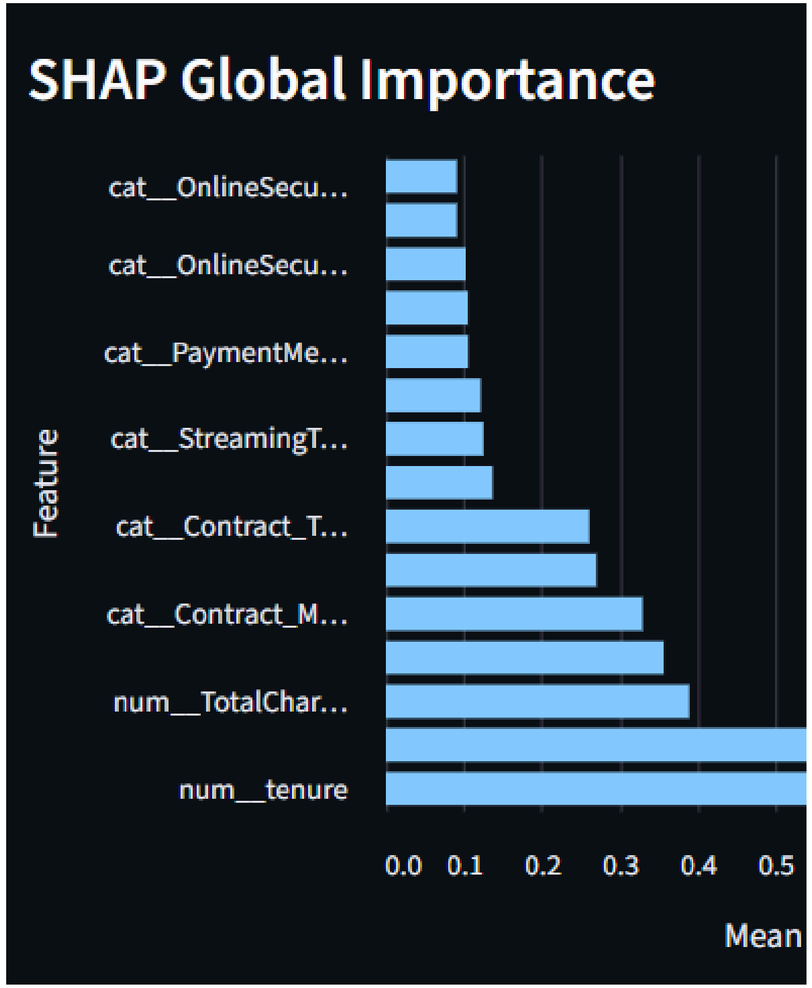
Why threshold matters

- Lower threshold → higher recall (catch more churners) but more false positives
- Higher threshold → higher precision (fewer wasted offers) but misses churners

Threshold scan (Precision / Recall)

- 0.2 → Precision 0.389, Recall 0.963 (very aggressive targeting)
- 0.5 → Precision 0.503, Recall 0.783
- 0.7 → Precision 0.602, Recall 0.602 (more conservative)

Takeaway: We should **not** pick 0.5 blindly — we need a business-optimized threshold.



Explainability

Why the Model Flags Churn?

Global explainability

- Permutation importance (business-friendly, raw feature impact)
- SHAP global importance (mean |SHAP| per feature)

Local explainability

SHAP waterfall explains one prediction:

- which features increased risk
- which reduced risk
- magnitude of each contribution

Why this matters

- Builds trust: stakeholders can understand drivers
- Enables action: focus retention efforts on known churn drivers (e.g., contract type, tenure, support services)

Business Simulation: ROI-Based Decision Policy

Core idea:

- Model outputs probabilities → business needs decisions.

Expected Value (EV) framework:

- EV per targeted customer:

$$EV = P(churn) \times save_rate \times (margin \times months_saved \times MonthlyCharges) - offer_cost$$

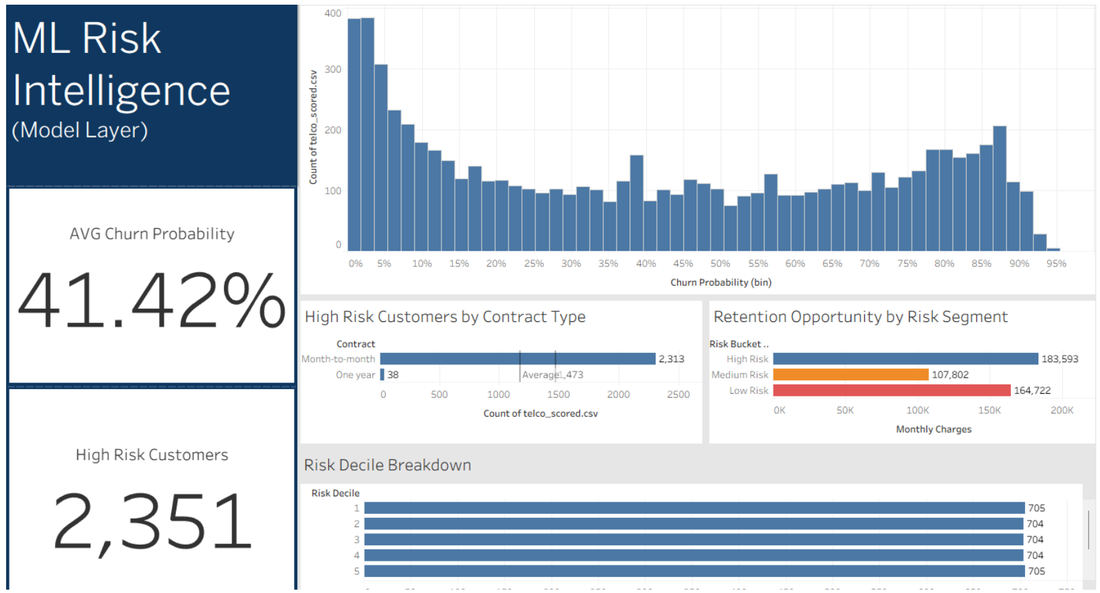
ROI simulation output:

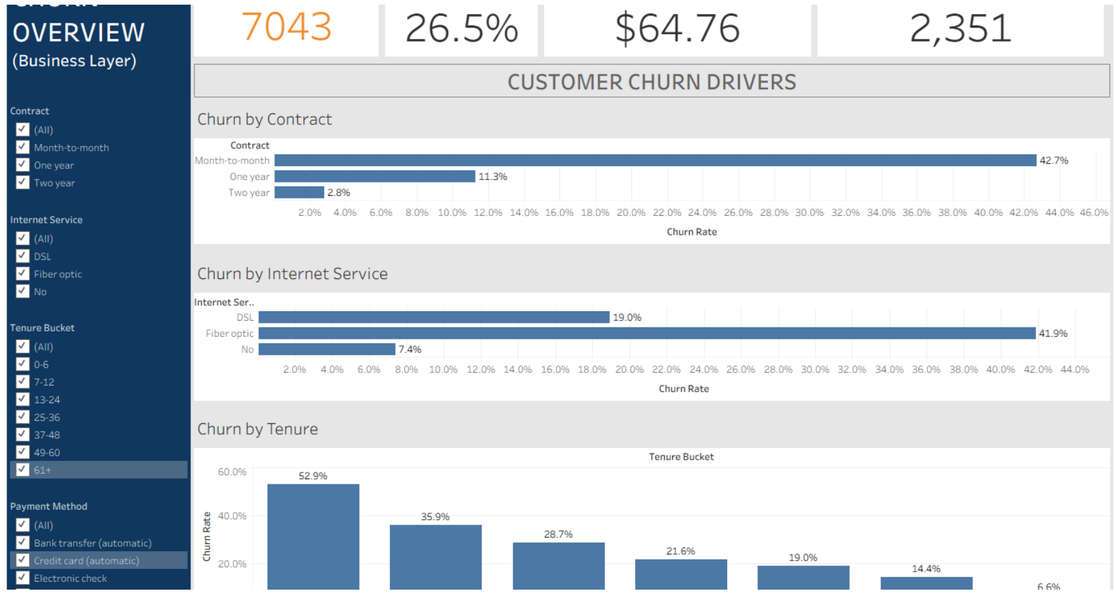
- Evaluated thresholds from ~0.10 to 0.90
- Selected threshold that maximizes total expected value

Best decision policy (from simulation):

- Best threshold: 0.15
- Targeted customers: 4992
- Total expected value: ~$198,995

Product + Dashboards + Monitoring (Real-World Delivery)





Streamlit (Modern SaaS UI):

- KPI cards + decision panel (Intervene vs No action)
- EV per customer estimate
- Analytics dashboard (stacked distribution, top drivers, precision/recall vs threshold)
- Batch scoring + prediction logs

Built from exported CSVs:

- telco_scored.csv (probabilities + risk buckets/deciles)
- roi_thresholds.csv (threshold → expected value)
- feature_importance.csv (top churn drivers)

Monitoring (Model lifecycle):

- Predictions logged → current dataset
- Evidently drift report compares reference vs current
- Output: data_drift_report.html

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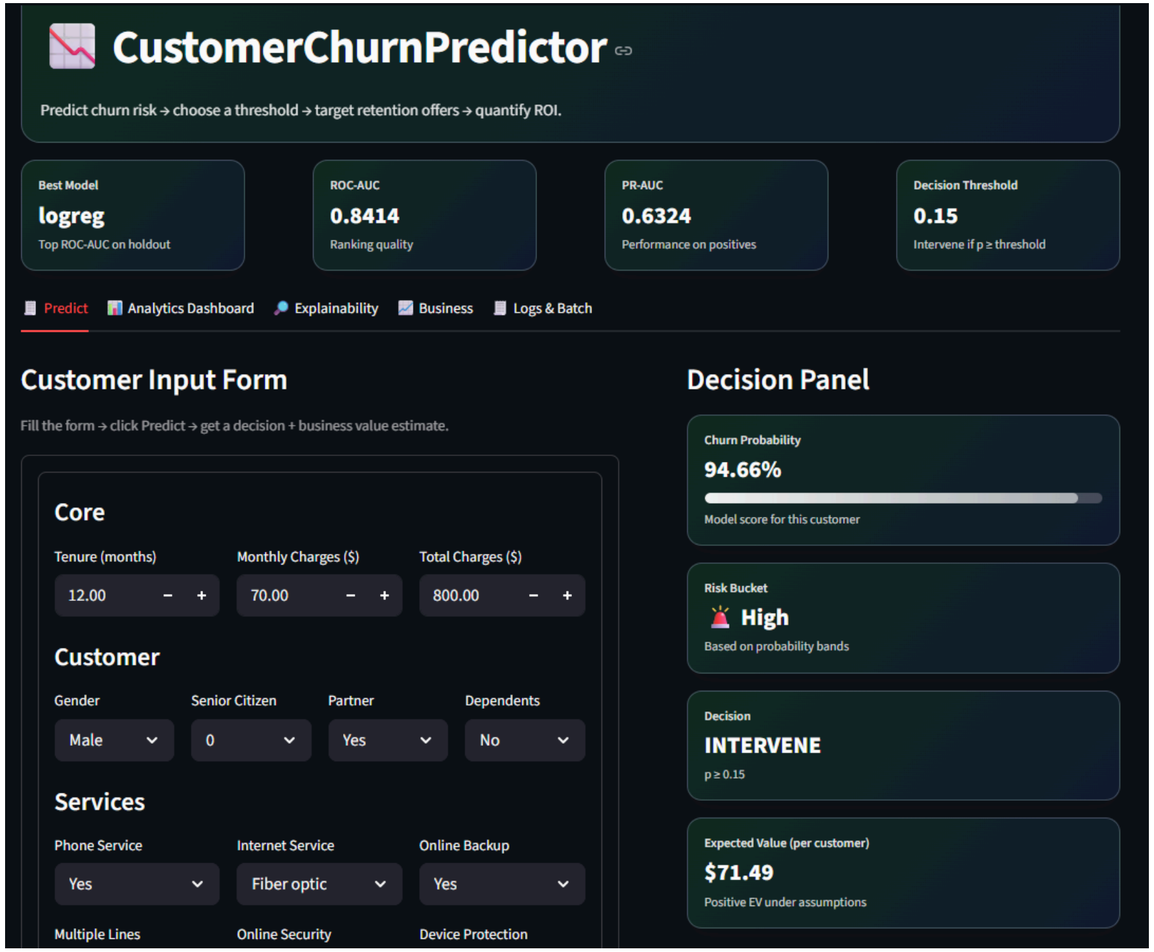

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